Xinqi Wang

(206) 512-5448 | wangxqkaxdd@gmail.com | elliotxinqiwang.github.io

Education

University of Washington (UW), Seattle, WA, USA Sep 2023 – Present

Ph.D. in Computer Science & Engineering, Paul G. Allen School of Computer Science & Engineering

Tsinghua University (THU), Beijing, China Aug 2019 – Jul 2023

B.Eng., Institute for Interdisciplinary Information Sciences (Yao Class)

Research Interests: Machine Learning; Reinforcement Learning; Learning Theory; Deep Learning; Agentic Evaluation; Automated Software Engineering.

Publications

*(**Bold** indicates author; * equal contribution.)*

[1] Lucas Jing, **Xinqi Wang**, Liao Zhang, Simon S. Du. "PBT-Bench: Benchmarking AI Agents on Property-Based Testing." Preprint, 2026. arXiv:2605.15229.

[2] **Xinqi Wang**, Hao Hu, Simon S. Du. "Policy-Based Trajectory Clustering in Offline Reinforcement Learning." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.

[3] Liao Zhang, Tong Chen, Xiwei Wu, Qi Liu, Xiyu Zhai, **Xinqi Wang**, Qinxiang Cao. "Neuro-Symbolic Generation and Validation of Memory-Aware Formal Function Specifications." Preprint, 2026. arXiv:2603.13414.

[4] **Xinqi Wang**, Natalia Zhang, Qiwen Cui, Runlong Zhou, Sham M. Kakade, Simon S. Du. "Preference-Based Multi-Agent Reinforcement Learning: Data Coverage and Algorithmic Techniques." 2024.

[5] Chuning Zhu, **Xinqi Wang**, Tyler Han, Simon S. Du, Abhishek Gupta. "Distributional Successor Features Enable Zero-Shot Policy Optimization." *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[6] **Xinqi Wang**, Qiwen Cui, Simon S. Du. "On Gap-dependent Bounds for Offline Reinforcement Learning." *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Experience

Research Intern, MiroMind Apr 2026 – Present

- Built data-cleaning and repair pipelines for large-scale coding datasets, improving the quality and reliability of training data for code-focused large language models.
- Designed reinforcement learning algorithms for training large language models on coding and reasoning tasks.
- Optimized the RL inference stack, accelerating rollout generation to improve end-to-end training throughput.

Research Experience

PBT-Bench: Evaluating AI Agents via Property-Based Testing in Large-Scale Repositories Oct 2025 – May 2026

Lead researcher; Advisor: Simon S. Du | *Released as a preprint (arXiv:2605.15229)*

- Conceptualized and developed PBT-Bench, a novel framework that fills the gap in assessing AI agents' ability to perform document-driven debugging and property-based verification within complex software ecosystems.

- Engineered an automated bug-injection pipeline that systematically introduces logical and functional regressions into production-grade Python packages, requiring agents to reconcile natural-language documentation with source code.
- Architected the evaluation infrastructure assessing whether agents generate robust, property-based test functions that expose injected bugs and ensure long-term code reliability.
- Curated a diverse dataset of 100 problems across 40 libraries with 365 injected bugs, establishing a rigorous "Agentic Debugging" metric that moves beyond code completion toward deep architectural understanding.

Fisher-LDA Driven Benchmark Distillation for LLM Reasoning Evaluation Apr 2025 – Sep 2025

Lead researcher; Advisor: Tianyi Li

- Proposed a generalized benchmark-distillation framework that optimizes LLM evaluation efficiency by identifying high-discriminative test cases across reasoning modes (CoT/Think vs. Direct/No-Think).
- Engineered a scoring metric based on question-wise performance differentials to quantify the sensitivity of benchmarks (AIME, GPQA, GSM8K) to internal reasoning.
- Implemented Fisher Linear Discriminant Analysis to weight individual test items, maximizing between-class variance of reasoning behaviors while minimizing within-class noise.
- Reduced redundant evaluation overhead by filtering non-discriminatory problems, producing a distilled subset that retains high alignment with full-scale benchmark results while lowering inference cost.

Policy-Guided K-means: A Clustering Approach for Offline RL Trajectories Oct 2024 – May 2025

Lead researcher; Advisor: Simon S. Du | *Accepted at UAI 2026*

- Designed a new algorithm, PG-Kmeans, for clustering offline-RL datasets.
- Derived theoretical convergence guarantees for the algorithm.
- Designed and ran experiments demonstrating PG-Kmeans' superiority over traditional VAE-based methods.

Offline Multi-Agent RL with Human Feedback Feb 2024 – May 2024

Lead researcher; Advisor: Simon S. Du

- Investigated existing MARL environments and built the required environment.
- Constructed the MARL environment database with MAPPO and VDN.
- Designed and implemented a code pipeline for offline multi-agent RL with human feedback, and verified the role of reward regularization in reward modeling.

Distributional Successor Features Enable Zero-Shot Policy Optimization Sep 2023 – Feb 2024

Lead researcher; Advisors: Simon S. Du, Abhishek Gupta

- Co-designed the algorithm for task-agnostic RL settings.
- Developed performance bounds for the proposed algorithm.
- Adapted experiments to verify the algorithm.

Extra-Agent Method in Multi-Agent Reinforcement Learning Jul 2022 – Feb 2023

Lead researcher; Advisor: Simon S. Du

- Analyzed the inherent defects of the naive Policy Gradient (PG) method in MARL.
- Unified prior PG-based improvements into two main categories.
- Proposed a new algorithm family with performance lower-bounded by PG and random search.

On Gap-dependent Bounds for Offline Reinforcement Learning Mar 2022 – Jun 2022

Lead researcher; Advisor: Simon S. Du | *Accepted at NeurIPS 2022*

- Analyzed algorithms for offline learning in tabular cases and provided matching upper and lower bounds on gap-dependent sample complexity.

- Introduced a new technique to analyze LCB-style algorithms.
- Proved the necessity of the overall optimal-policy coverage assumption for LCB.

Causal Relationships in Autonomous Driving Jul 2021 – Aug 2021

Advisor: Yang Gao (IIIS, THU)

- Ran experiments on the copycat problem across autonomous-driving environments with different imitation-learning algorithms, and analyzed the results.

Skills

Programming	C, Python, PyTorch, MATLAB, LaTeX
Languages	TOEFL iBT 106 (R30/L28/S23/W25); GRE 328+3.5 (V158/Q170); JLPT N1 (highest level).